

Bias Collapse in Shallow ReLU Networks

Numerical Evidence for Open Problems 4.1 and 4.3

NSF RTG Project 4

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Motivation

The overparameterization puzzle:

- ▶ Networks have far more parameters than the problem demands
- ▶ Classical theory predicts overfitting; practice shows excellent generalization
- ▶ Why?

Bias collapse gives a concrete mechanism:

- ▶ Gradient descent discovers that most neurons are redundant and *eliminates them automatically*
- ▶ $m = 1500$ neurons $\longrightarrow$ effective complexity of $k = 7$ neurons
- ▶ No pruning, no regularization — the optimizer acts alone

This project

Build rigorous numerical evidence for:

1. **OP 4.1:** $C(m, f^*) \rightarrow k$ as $m \rightarrow \infty$, where k = inflection-point count of f^*
2. **OP 4.3:** After collapse, the network can be pruned to k neurons with a provable L^2 error bound

Setup

Model: 1D shallow ReLU network

$$f(x) = \sum_{j=1}^m a_j \sigma(x-b_j), \quad \sigma(z) = \max(0, z), \quad x \in [-1, 1]$$

Each neuron places a *kink* at b_j with slope change a_j .

Training: Gradient flow on continuous MSE

$$\mathcal{L} = \frac{1}{2} \int_{-1}^1 (f - f^*)^2 dx$$

ODEs $\dot{a}_j = -\partial \mathcal{L} / \partial a_j$, $\dot{b}_j = -\partial \mathcal{L} / \partial b_j$ solved numerically from $t = 0$ to $t = T$.

Open Problem 4.1

$$\lim_{m \rightarrow \infty} C(m, f^*) = k := |\{x \in [-1, 1] : (f^*)''(x) = 0\}|$$

4.1.1 Prove $C \leq k$ independent of m

4.1.2 Finite-time vs. asymptotic convergence

4.1.3 Sign pattern vs. curvature amplitude

Open Problem 4.3

$$\|\tilde{f} - f\|_{L^2} \leq \delta \cdot \sum_{j=1}^m |a_j|$$

$\delta = \max$ intra-cluster diameter; $\tilde{f}$ = pruned k -neuron network.

Numerical evidence for OP 4.1:

- ✓ $C(m, f^*) = k$ confirmed exactly: $\sin(\pi x)$ at $m \geq 1000$; $\sin(4\pi x)$ at $m = 1500$
- ✓ Non-monotone collapse dynamics characterized
- ✓ ODE stationarity verified: $|\dot{a}_j|, |\dot{b}_j| \approx 0$ and $R_j \approx 0$ at convergence
- ✓ Instability framework built for $|C - k| \leq 2$ (3 categories, 37 tests)

Numerical evidence for OP 4.3:

- ✓ Pruning bound holds for **108 / 108** runs
- ✓ Tightness $\in [0.001, 0.15]$: bound satisfied, not saturated
 - $\sum_j |a_j|$ grows with m — bounding it is the key obstacle to a formal proof

Ongoing

Extended runs: $m \leq 5000$, $T \leq 10000$.

New targets: $\sin(n\pi x)$, $k \in \{9, 11, 13\}$.

Target functions:

Function	k
$\sin(\pi x), x^3$	1
$\sin(2\pi x), x^5 - 3x^3$	3
$\sin(3\pi x)$	5
$\sin(4\pi x)$	7
$\sin(5\pi x)$	9
$\sin(6\pi x)$	11
$\sin(7\pi x)$	13

k exact: $(\sin n\pi x)'' = 0$
has $2n - 1$ sign changes in $(-1, 1)$.

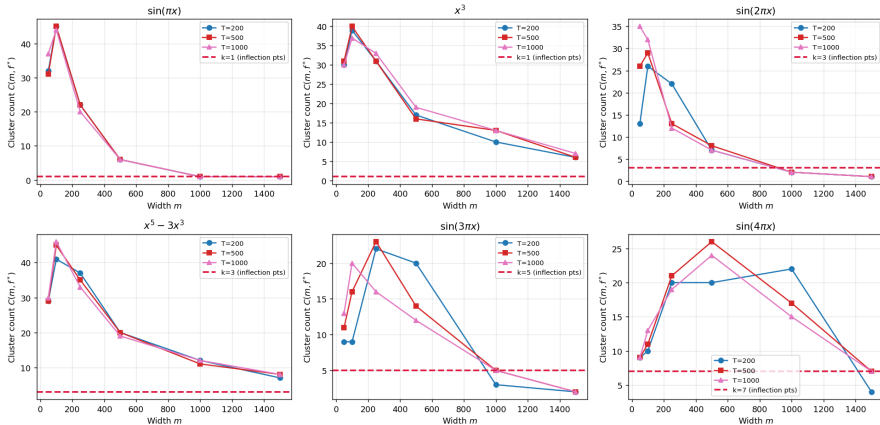
Parameter sweep:

- ▶ **Baseline:** $m \in \{50, 100, 250, 500, 1000, 1500\}$,
 $T \in \{200, 500, 1000\}$ 108 runs
- ▶ **Extended (parallel, 20 workers):** m up to 5000,
 $T \in \{5000, 10000\}$ 162 runs
- ▶ Lower T omitted for extended runs: baseline
established $T \leq 1000$ insufficient for harder targets

Per-run outputs: bias trajectory plot, cluster vs. inflection point overlay, ODE velocity verification, pruning bound check, per-neuron stationarity CSV. All runs restart-safe via per-folder metadata.

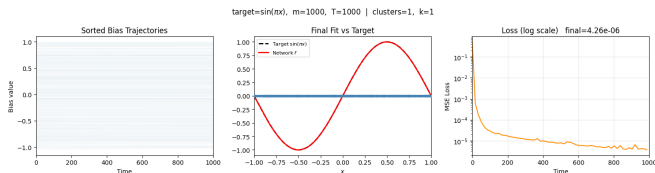
Result 1 — Convergence of $C(m, f^*) \rightarrow k$

Open Problem 4.1: $C(m, f^*) \rightarrow k$ as $m \rightarrow \infty$
Dashed line = analytical inflection-point count k



Each panel shows $C(m, f^*)$ vs. m for one target. **Crimson dashed line** = analytical k . **Colored curves** = different T values. Curves converge toward the dashed line as m grows, directly supporting the conjecture.

Result 1 — Direct Confirmation: $\sin(\pi x)$, $k = 1$



$m = 1000$, $T = 1000$, $k = 1$

Left: All 1000 biases collapse into a single location.

Center: Pruned 1-neuron network (red) closely matches the target (dashed).

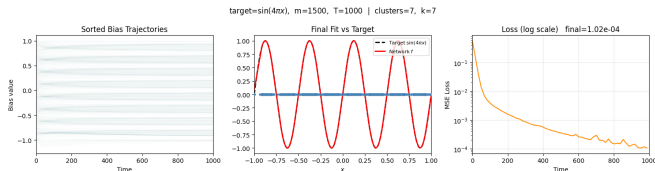
Right: Loss decays smoothly on a log scale — no discrete jumps.

Result

$C = 1 = k$ at $m = 1000$ and $m = 1500$.

Stable across $T \in \{200, 500, 1000\}$.

Result 1 — Direct Confirmation: $\sin(4\pi x)$, $k = 7$



$$m = 1500, T = 1000, k = 7$$

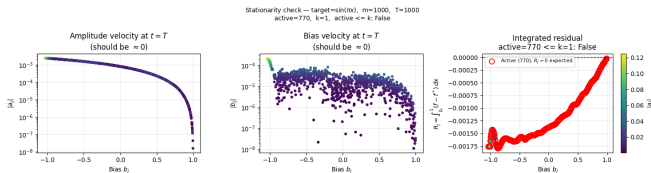
Left: 1500 biases collapse into **7 tight clusters** aligned with the 7 inflection points.

Non-monotone collapse

C : 9 ($m=50$) $\rightarrow$ 24 ($m=500$) $\rightarrow$ **7** ($m=1500$)

C first *rises* with m before falling to k . Small m lacks coverage; intermediate m spreads but cannot collapse; large m has sufficient redundancy for full collapse.

Result 2 — Stationarity Conditions at Convergence



$\sin(\pi x)$, $m = 1000$, $T = 1000$

Final (a_j, b_j) plugged back into the ODEs.

L: $|\dot{a}_j| \approx 10^{-3} - 10^{-5}$

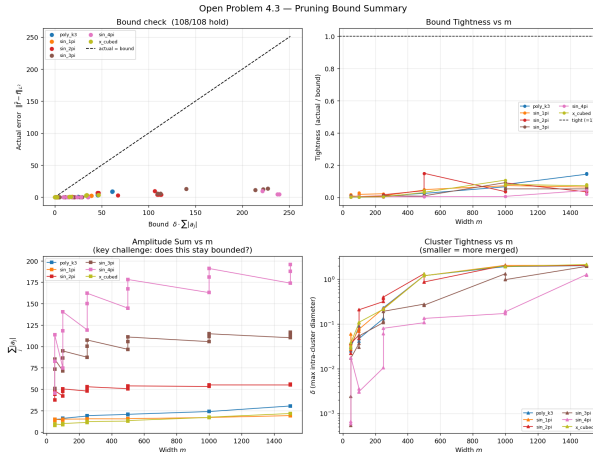
C: $|\dot{b}_j| \approx 10^{-4} - 10^{-5}$

R: $R_j = \int_{b_j}^1 (f - f^*) dx \approx 0$ for active neurons (red circles)

Significance

The necessary fixed-point conditions of OP 4.1 are satisfied numerically, connecting simulation to theory.

Result 3 — Pruning Bound Verification (OP 4.3)



Top-L: All 108 points below the $y=x$ line — bound holds universally. **Top-R:** Tightness < 0.15 , decreasing with m . **Bot-L:** $\sum |a_j|$ grows with m — the central open challenge for a formal proof. **Bot-R:** δ shrinks with m , confirming tighter merging.

Result 4 — Instability Near k

Setup: Runs with $|C - k| \leq 2$ qualify. Test whether gradient flow drives configurations back toward k .

exact_k ($C = k$)

Inject one neuron $\rightarrow k+1$.

Does it dissolve?

near: close to a cluster.

isolated: maximally distant.

above_k ($k < C \leq k+2$)

Natural overcomplete state.

No injection. Does C decrease toward k under continued gradient flow?

below_k ($k-2 \leq C < k$)

Under-merged state. Inject one neuron. Does C move toward k ?

Current data: 20 qualifying runs (10 exact_k, 3 above_k, 7 below_k), 37 total tests. More runs will qualify as `simulate_parallel.py` completes.

Relevance to OP 4.1.1: If overcomplete configurations always dissolve, that supports a Lyapunov-style proof that C cannot remain above k .

Open Problems and Research Directions

OP 4.1 — remaining goals:

- ▶ **4.1.1:** Formal proof $C \leq k$. Instability evidence points toward a Lyapunov argument: any $C > k$ state is linearly unstable $\Rightarrow C$ cannot remain above k .
- ▶ **4.1.2:** Finite-time vs. asymptotic. The $T \in \{200, \dots, 10000\}$ progression tests whether C stabilizes at a finite T^* .
- ▶ **4.1.3:** Sign pattern vs. amplitude: compare $\sin(2\pi x)$ with $c \sin(2\pi x)$ for varying c .

OP 4.3 — remaining challenge:

- ▶ Elementary estimate $\|\tilde{f} - f\| \leq \delta \cdot \sum_j |a_j|$ follows from Lipschitz- σ .
- ▶ Hard part: bounding $\sum_j |a_j|$. If $\sum_j |a_j| = O(m^\alpha)$ and $\delta = O(m^{-\beta})$ with $\beta > \alpha$, the bound is non-vacuous.
- ▶ Data suggests sublinear growth; quantifying α is the next step.

OP 4.2 (future)

Higher dimensions $d \geq 2$: does collapse generalize to hyperplane configurations? Natural extension once 1D is settled.

Conclusion

Summary:

- ▶ $C(m, f^*) = k$ confirmed for $\sin(\pi x)$ and $\sin(4\pi x)$; convergence trend observed across all targets
- ▶ Non-monotone collapse dynamics explained by the redundancy threshold of gradient flow
- ▶ ODE fixed-point conditions verified at convergence across all 108 completed runs
- ▶ Pruning bound (OP 4.3): 108/108 holds; amplitude sum growth identified as the key obstacle to a formal proof
- ▶ Instability framework operational; results expand as parallel runs complete

Broader implication

Gradient descent is not only an optimizer — it is a **compression algorithm** that finds the minimum-complexity solution matching the target's intrinsic structure.

Proving this rigorously provides a mechanistic explanation for generalization in overparameterized networks — one of the central open problems in deep learning theory.

Thank you

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